

Research Statement

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My research to date spans three main areas: Bayesian inference for spatial statistics, surrogate modeling for computer experiments, and inferential models for decision making theory. The first two projects were motivated by real-world applications and contributed to the fields by providing novel methods for multiple outcomes and multiple computer experiments. The last project, currently ongoing, develops decision-making theory using a framework outside the likelihood-based methods and Bayesian framework. I found that both applications and fundamentals have been guiding my research interests and will continue to guide my career.

1 Spatial and Bayesian

Pollutants, such as Per- and Polyfluoroalkyl Substances (PFAS), are often co-existing, correlated, and causing issues on multiple conditions. Furthermore, it is often difficult to know that all confounders have been included in the model. Using causal inference in the context of spatial data has received more attention (Reich et al., 2021). These reasons motivated my advisor, Dr. Brian Reich, his former postdoctoral fellow, Dr. Yawen Guan, and me to develop a multivariate spectral model for multiple exposures and multiple outcomes. Originated from Guan et al. (2023), the method first projects the spatial information into the spectral domain, making information independent across frequencies and facilitating fast computation. Spectral frequencies are analogous to spatial scales; high frequencies correspond to local scales, while low frequencies correspond to global scales.

Two challenges emerge: 1) unmeasured spatial confounding is likely in environmental applications. Thus, we relax the no-unmeasured-confounder assumption and allow for global confounding. This means the coefficient estimates should be multiscale, so that the estimate corresponding to local scales are unbiased. This, however, leads to the second challenge: 2) the coefficient estimates are in a three-way tensor and high-dimensional. We utilize canonical polyadic (CP) tensor decomposition to reduce dimension, ensuring efficient computation. I constructed the fully Bayesian model with several MCMC samplers in R, and the manuscript was submitted in June 2025 and is currently under review (Prim et al., 2025).

1.1 Future directions

I hope to continue the work in spatial statistics and examine the issue of unmeasured spatial confounding. Methods that account for unmeasured spatial confounding sometimes seem to lead to contradictory results (Dupont et al., 2024). The role played by spatial scales is particularly interesting; confounding can vary at different scales and paying close attention to the scales can help disentangle different degrees of confounding (Paciorek, 2010).

Further, my experience in spatial analysis can potentially transfer to image analysis, e.g. machine learning for MRI imaging analysis. I will seek collaboration opportunities with medical researchers, such

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as Mike Chan, Professor of Radiation Oncology at Wake Forest University School of Medicine, with whom I have previously collaborated.

2 Surrogate modeling for computer experiment

My second research project started in my 2024 summer internship at the Lawrence Livermore National Laboratory. Constructing an aerodynamic database, which describes conditions for flights—speed, altitude, angles of the aircraft, angles of fins, forces, and moments—can be computationally expensive, and surrogate modeling, often with Gaussian processes, can greatly reduce the computational cost (Gramacy, 2020; Sauer, 2023). Further, if the goal is to construct a trimmed database, which only contains flying conditions of interest, instead of a brute force and computationally expensive approach—using computer experiments to arrive at input/output values on a dense grid and only retain the conditions of interest—active learning and contour location methods can save cost (Quinlan et al., 2024). In short, an algorithm is needed to efficiently find an optimal design point that leads to multiple responses that meet a certain criterion. Specifically, let $f^{(r)} : \mathcal{X} \rightarrow \mathbb{R}$ denote the r^{th} computer experiment for $r = 1, \dots, R$. Each $f^{(r)}$ is independent but acts on the same domain $\mathcal{X} \subset \mathbb{R}^d$. Our goal is to identify the “optimal design point”:

$$\left\{ \tilde{\mathbf{x}} \in \mathcal{X} \mid f^{(r)}(\tilde{\mathbf{x}}) = \tau_r \ \forall \ r = 1, \dots, R \right\}, \quad (1)$$

where τ_r are pre-specified target response values.

Motivated by this example, my mentor at LLNL, Dr. Kevin Quinlan, and I started the internship with a literature review on surrogate modeling, active learning, and contour location methods during the internship. After the summer, I stayed on as a collaborator and we invited Dr. Annie Booth, a NCSU faculty at that time but currently at Virginia Tech who has worked extensively in this particular area, to join us to develop a method for efficiently construct a trimmed database. One major challenge is that current contour location methods are for one single computer experiment (Booth et al., 2025; Cole et al., 2023). Since then, we developed a method that use joint predictive probabilities to target the intersection of multiple contours. When necessary, we sample points from the Pareto front to explore areas with high uncertainty. Our method—joint Contour Location (jCL)—balances exploitation and exploration and outperforms competitors in both simulations and a real-world application. We are currently completing the manuscript (Prim et al., 2025+) and planning to submit the paper and upload a preprint this November.

2.1 Future directions

After the paper is submitted, I plan to package the Python code into a Python package, allowing for easy installation and use. Besides, the setting for the real-world application is simplified such that one and exactly one optimal design point exists at the intersection of multiple contours. We hope to explore more difficult scenarios, in which no or multiple optimal design points exist.

3 Valid inference

While developing methods for tangible applications is exciting, I am also deeply interested in fundamentals and theoretical guarantees of validity. IM provides uncertainty quantification and guarantees error rates to be below a prespecified level (Martin, 2025b). Based on possibilistic measures, IM incorporates imprecise probability; analogous to integration for probability measures, IM uses optimization and often requires more expensive computation. Some new strategies based on Monte Carlo sampling have been proposed and opened up exciting opportunities (Cella and Martin, 2025; Martin, 2025a).

With my two other co-advisors, Dr. Ryan Martin and Dr. Jonathan Williams, my third research project—currently ongoing—explores how to apply Inferential Models (IM) on decision making theory. We are examining different examples to demonstrate the strengths of IM and exploring efficient computation strategies and demonstrating decision-making using IM, striving to better explain how IM can offer a valid and attainable option for statistical method. Our goal is to complete an invited paper for the International Journal of Approximate Reasoning (IJAR) in December 2025 and continue exploring the utility of IM on decision-making beyond this year.

3.1 Future directions

After the completion of the current decision-making project, I hope to continue working with Ryan Martin and Jonathan Williams and potentially Leo Cella at Wake Forest University on other topics in valid inference and inferential models.

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